MOHAMMAD ZIRDI SYUKUR

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EDUCATION

University of Oxford, UK *September 2026 - In progress*
MSc in Quantum Technologies

Yale-NUS College, Singapore *August 2019 - June 2023*
BSc Honours in Physical Sciences (Cum laude)

WORK EXPERIENCE

Research Engineer, A*STAR, Q.InC Singapore *July 2023 - July 2026*

- Worked with Dr. Junyi Lee in the atomic magnetometry group to build quantum vapour sensors for medicine, navigation, information, and geophysical exploration technologies
- Research responsibilities: atomic and optical experiments, CAD, data analysis and programming
- Designated procurement associate, and risk management / safety officer

PROJECTS

QEP 2.0 Atomic Magnetometer *July 2023 - October 2024*

- Built and tested a prototypical atomic magnetometer capable of part per billion sensitivity measurements within earth's field. I designed, simulated, and fabricated various, optical-mechanical mounts, gradient coils, laser systems, temperature control loops, and data acquisition/analysis

HTX NQR-based Radio Frequency Magnetometer *July 2025 - July 2026*

- Developing NQR-based unexploded ordnance and crystalline-narcotics detector for counter-terrorism and import control to be deployed at national borders.
- Designed the optical pump lasers, optomechanics, electromagnets, and oven

Hybrid Pumping Investigation *January 2025 - July 2026*

- Experiment that challenges assumptions in literature about spin-temperature equilibrium in hybrid vapour cells. Intended to improve vapour cell uniformity and hence magnetometer sensitivity

Low Noise Electronics *July 2023 - Nov 2024*

- Built custom low-noise current sources and amplifiers used for stable magnetometry references.
- Achieved near-equivalent low-noise to (often unavailable) commercial products in the RF regime

Gradient-coil Electromagnet for Ultra-Cold Atoms *December 2020 - May 2023*

- Optimized electromagnet design by programming magnetic field simulations in Julia. Intended for use with ultra-cold atoms to tune atomic interaction strength in a quantum simulation experiment
- Became my undergraduate thesis under supervision of Ben A. Olsen

PROGRAMMING SKILLS :

Python, Git, LaTeX, Julia, R, Java, Bash, HTML, CSS

OUTREACH

- Chair of **Quantum Young Researcher's Association** (2024-2026), a grass-roots forum of 300 young Singaporean professionals in the field of quantum. We facilitated their conversations with national-level policy makers, industries, and universities to discuss research objectives, ethics, politics for quantum in Singapore and Southeast Asia
- Representative in the **International Year of Quantum, Singapore** (2024-2025), to celebrate UNESCO's 100 years of Quantum, we organized year-long outreach events for quantum science at Singaporean museums, conferences, and schools.